

Practical 9

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

```
GNU nano 8.7
#include<stdio.h>

int main()
{
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int n=12;
    int i,j,k;
    int hits=0,faults=0;

    for(i=0;i<3;i++)
        frames[i]=-1;

    for(i=0;i<n;i++)
    {
        int found=0;

        for(j=0;j<3;j++)
        {
            if(frames[j]==pages[i])
            {
                hits++;
                found=1;
                break;
            }
        }

        if(found==0)
        {
            int pos=0,farthest=i;

            for(j=0;j<3;j++)
            {
                for(k=i+1;k<n;k++)
                {
```

```
GNU nano 8.7 optimal.c
}

if(found==0)
{
    int pos=0,farthest=i;
    for(j=0;j<3;j++)
    {
        for(k=i+1;k<n;k++)
        {
            if(frames[j]==pages[k])
                break;
        }
        if(k>farthest)
        {
            farthest=k;
            pos=j;
        }
    }
    frames[pos]=pages[i];
    faults++;
}

printf("Page Hits = %d\n",hits);
printf("Page Faults = %d\n",faults);

printf("Hit Percentage = %.2f%%\n",(hits*100.0)/n);
printf("Fault Percentage = %.2f%%\n",(faults*100.0)/n);

return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execut
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justif
```

```
De11@aryaaa MSYS ~
$ nano optimal.c

De11@aryaaa MSYS ~
$ gcc optimal.c -o optimal
./optimal
Page Hits = 5
Page Faults = 7
Hit Percentage = 41.67%
Fault Percentage = 58.33%

De11@aryaaa MSYS ~
$
```